

Logic hazards

Combinational logic can glitch when inputs change

One input moves, paths differ

- Hazard analysis usually changes one input at a time
- Different path delays let one SOP term drop before another picks up
- A consensus term that bridges adjacent cubes, like BC here, can cover the gap
- Equal delays can hide the glitch in simulation but not in silicon

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Static / dynamic hazards

Single-input transitions on multi-level logic can glitch. Spot **static-1**, **static-0**, and **dynamic** hazards — and cover them.

Starter example: $F = AB + A'C$, flip A: $1 \rightarrow 0$ with $B=C=1$. Term AB falls before A'C rises $\rightarrow$ **static-1** glitch. Cover with consensus + BC.

Load starter example

Challenges (1/22)

Quiz: **static-1**. Stay-1 with a dip to 0 is a? Answer: static -1

Your answer

Show hintCheckNextIdle

quiz-static1quiz-static0quiz-dynamicquiz-consensusstarter-loadsee-static1cover-applycover-safe

quiz-termstatic0-presetsee-static0dynamic-presetsee-dynamicquiz-singleexplaindelay-gap

equal-delayquiz-sopquiz-posflip-holdrun-btnfull-fix

Core ideas

Static-1

Output should stay 1, briefly goes 0.

Static-0

Output should stay 0, briefly goes 1.

Dynamic

More than one edge when the function has one.

Logic hazards starter

learn_digital — logic hazards

Workbook practice

- In the workbook track
- Write the consensus term BC by hand
- Name static-one versus static-zero from the dip or spike direction
- Note one pitfall: assuming equal gate delays means no hazard in real hardware

Pitfalls to watch

- Do not confuse a hazard with wrong logic
- Cover terms must actually bridge the risky transition
- And remember: the browser lab is literacy
- Real designs still need timing analysis

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, identify one static hazard and one consensus fix
- When you are ready, take the short quiz, then continue to gate composer